

ONLINE SHOPPERS

Conversion Analysis Report

What helps online visitors become buyers?

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Executive Summary

This project studies **12,330 online shopping sessions**. The goal is to understand which behaviours are linked to a purchase and how the website can improve its conversion rate.

12,330	1,908	15.47%	85.57%
Total sessions	Purchases	Conversion rate	Returning visitor share

Main message: engagement matters. Visitors who buy explore more product pages and leave the website less quickly. The company should improve product discovery, reduce early exits and better support returning visitors.

In simple words

The website does not only need more visitors. It needs visitors who stay, explore products and continue until the purchase step.

Simple definitions

Session: one visit to the website.

Purchase conversion: a session that ends in a purchase.

Conversion rate: the percentage of sessions that end in a purchase.

Exit rate: how often visitors leave after viewing a page.

Returning visitor: a visitor who has already visited the website before.

1. Overall Conversion Performance

Out of every 100 sessions, about 15 end in a purchase and about 85 do not.

Most sessions do not end in a purchase

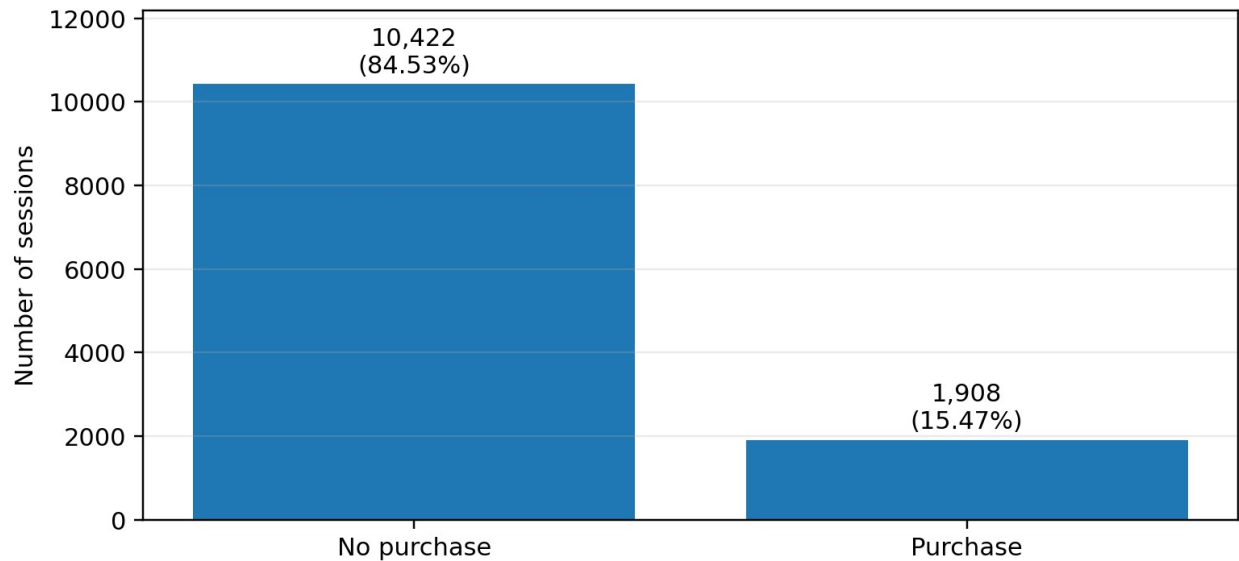


Figure 1. Purchase and non-purchase sessions.

What this means

The largest opportunity is inside the non-purchase group. The company should understand where these visitors stop and make the journey easier.

Key result: 1,908 sessions ended in a purchase, giving an overall conversion rate of 15.47%.

Dataset source: [UCI Machine Learning Repository - Online Shoppers Purchasing Intention Dataset](#)

2. Product Engagement

Visitors who buy spend more time exploring products. This is one of the clearest differences in the data.

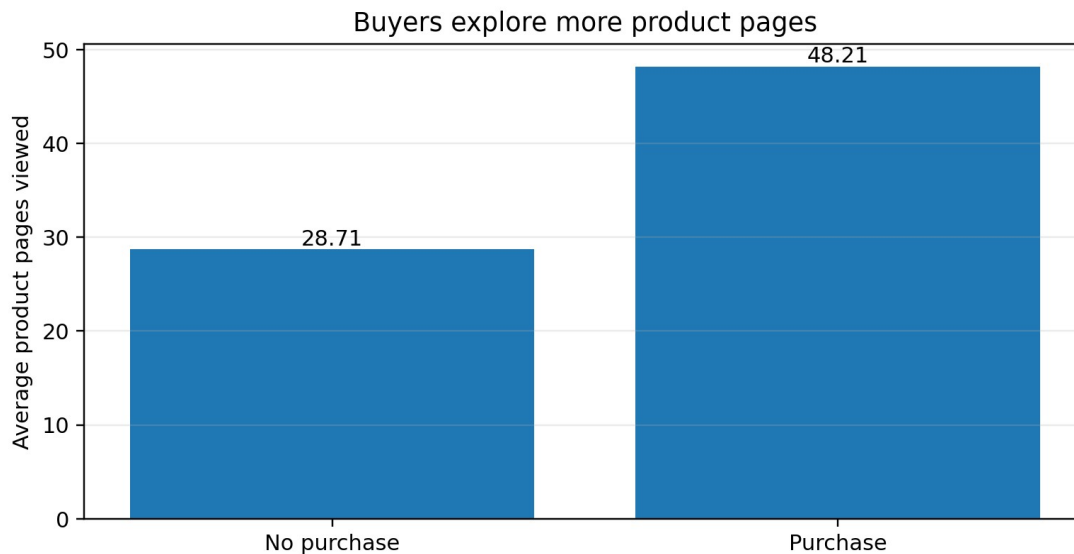


Figure 2. Average product pages viewed by purchase status.

Buyers view about **19.5 more product pages** than visitors who do not buy.

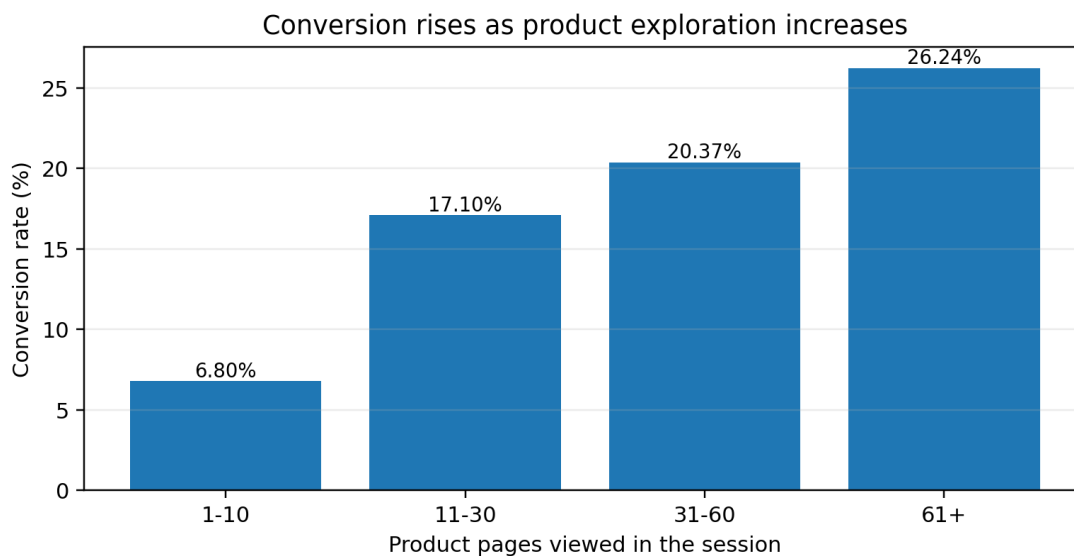


Figure 3. Conversion rate by number of product pages viewed.

What this means

Visitors who explore more products are more likely to buy. The website should make it easy to discover related products, compare options and move between categories.

3. Exit Behaviour

The exit rate shows how quickly visitors leave. A low exit rate means visitors continue through the website. A high exit rate means many visitors stop early.

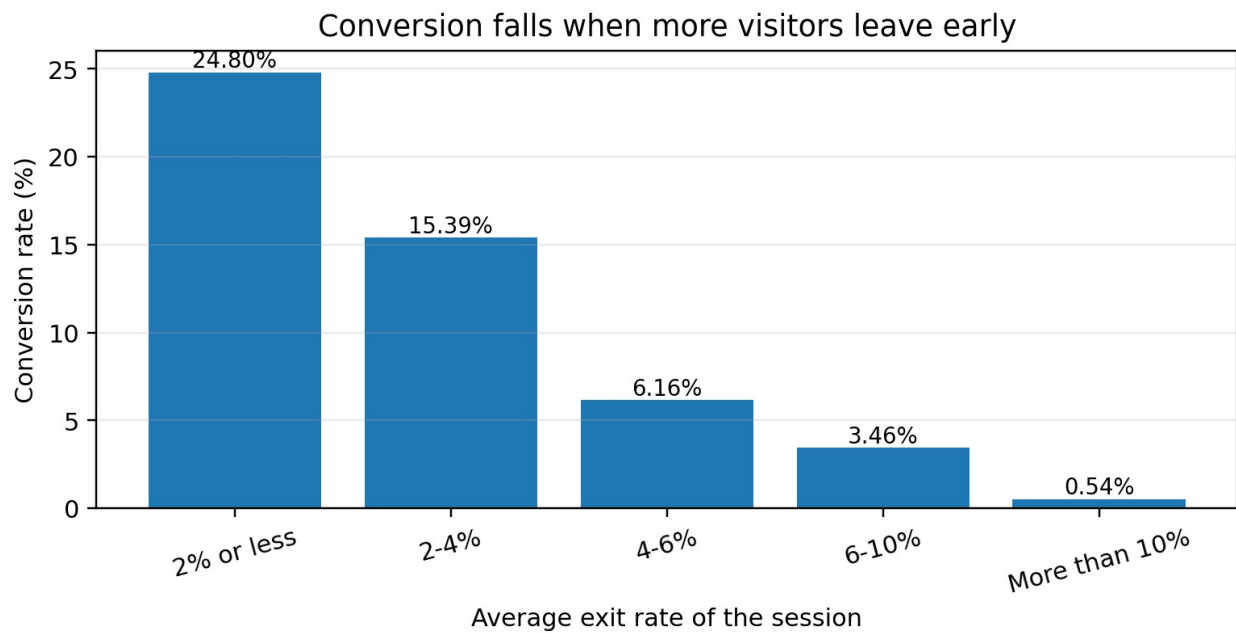


Figure 4. Conversion rate by exit-rate group.

The difference is very large: sessions with an exit rate of 2% or less convert at 24.80%, while sessions above 10% convert at only 0.54%.

What this means

Reducing early exits should be a priority. The company can test faster pages, clearer product information, stronger buttons and a simpler checkout.

4. Visitor Type and Timing

Different visitor groups do not convert in the same way.

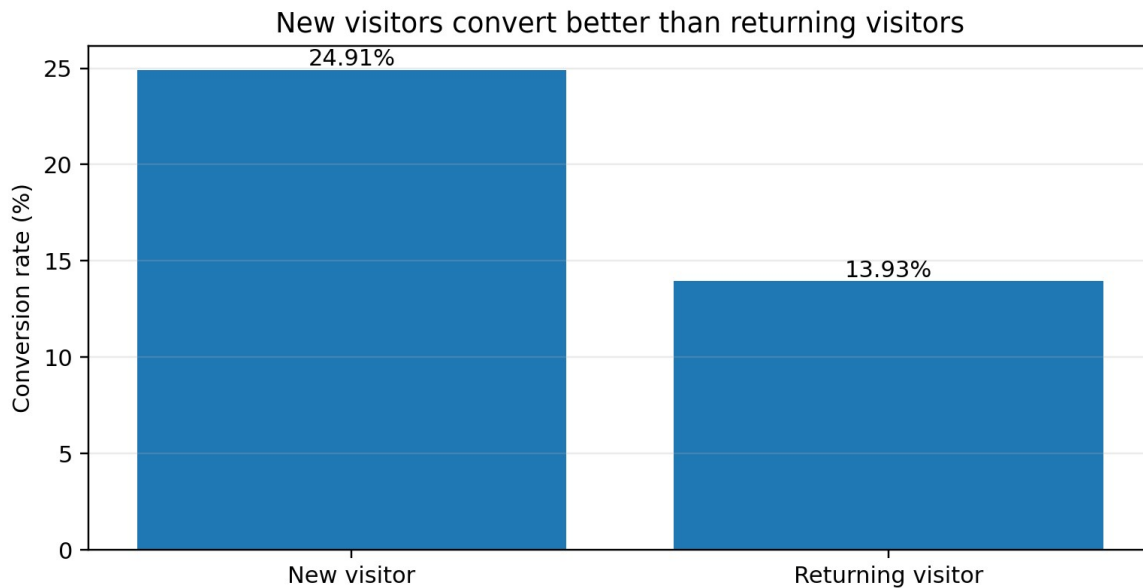


Figure 5. Conversion rate for new and returning visitors.

Important point: returning visitors represent most sessions, but their conversion rate is lower. This suggests that many people come back without completing the purchase.

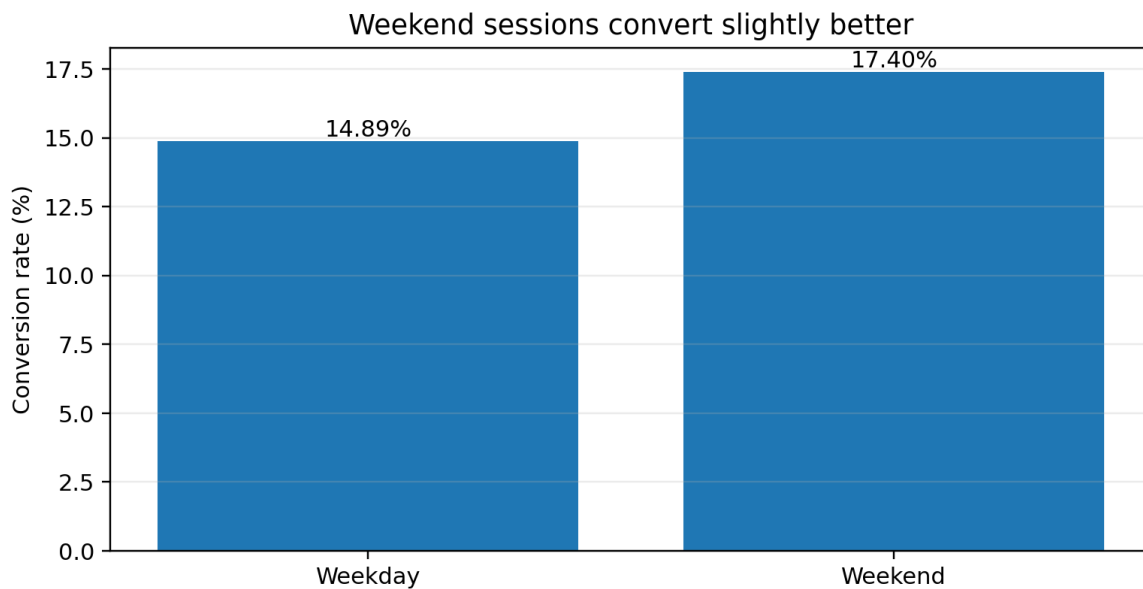


Figure 6. Weekday and weekend conversion rates.

What this means

The company should use saved carts, personalised reminders and return offers. Weekend campaigns may also work well because weekend conversion is slightly higher.

5. Seasonality and Traffic Quality

Conversion changes during the year. It also changes depending on the traffic source.

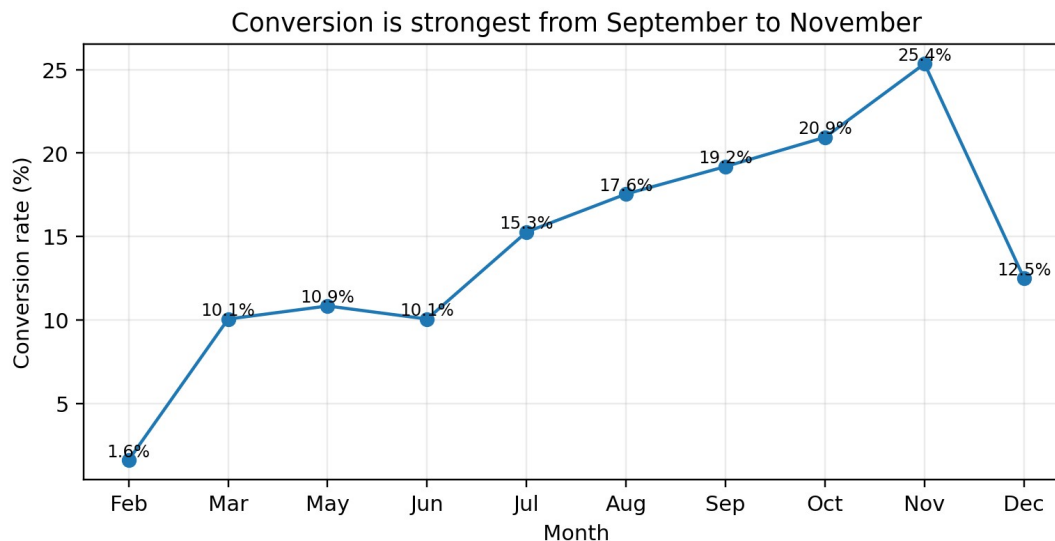


Figure 7. Monthly conversion rate.

Strongest period: September to November. November has the highest conversion rate at 25.35%.

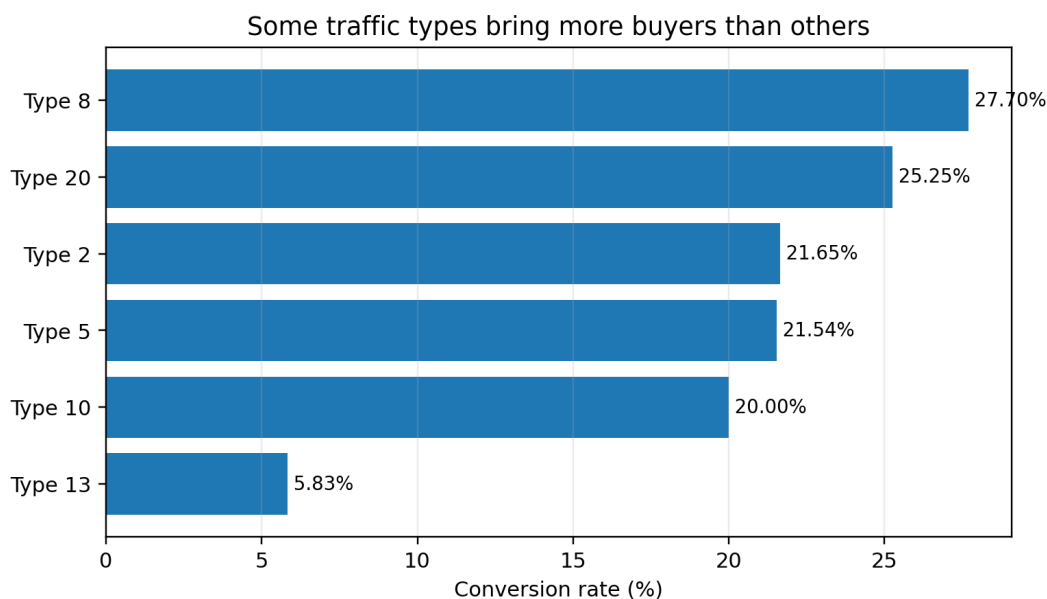


Figure 8. Selected traffic types with at least 100 sessions.

Important limitation: the traffic codes are anonymous. The company must connect Type 8, Type 20, Type 2 and the other codes to the real marketing channels before moving budget.

What this means

A channel with high traffic is not always the best channel. The company should compare both traffic volume and purchase conversion.

6. Business Recommendations

1. Reduce early exits

Improve page speed, product information, navigation, calls to action and checkout simplicity.

2. Help visitors explore products

Add related products, recommendations, better filters and comparison tools.

3. Support returning visitors

Use saved carts, personalised reminders and targeted offers for visitors who come back.

4. Prepare for strong months

Increase campaigns, stock and operational support before September, October and November.

5. Review traffic sources

Match the anonymous traffic codes to real channels and compare conversion with acquisition cost.

Recommended priority order

- First: fix pages and steps where visitors leave early.
- Second: improve product discovery and returning-visitor experience.
- Third: use seasonal and traffic insights to improve marketing decisions.

7. Limits and Final Conclusion

This analysis gives useful patterns, but the dataset does not contain everything needed for a complete customer analysis.

What is missing

- No customer ID: we cannot follow the same person across several sessions.
- No order value: we cannot calculate average order value or total revenue.
- No product details: we cannot identify which products drive or block conversion.
- No real traffic-channel names: the source codes cannot be linked directly to Google, email or social media.
- No marketing cost: we cannot calculate marketing return on investment.

How to read the results

The results show associations, not proof of cause. For example, visitors who view more products buy more often, but this does not prove that forcing people to open more pages will create purchases. Business actions should be tested with A/B tests.

Final conclusion

The company should not focus only on getting more traffic. It should attract qualified visitors, help them explore products and reduce the number of visitors who leave before buying.

Report based on: the Power BI dashboard pages Executive Overview, Engagement & Conversion, Traffic & Timing, and Business Recommendations, together with the cleaned project dataset.